CS260 | Data Structures
Reading Assignment - Week 1

1 Instructions

- This assignment is worth 5 points.

- **Read** "Chapter 2: Asymptotic Notation" from Algorithms Illuminated, by Tim Roughgarden. Chapter name is the same in both Part 1 and Omnibus edition versions of the book. (Note: This book has two versions. The old version was consisting of four parts whereas the new (omnibus) version has all in a single book. However, the content of the books are the same. So, I will be giving chapter names in reading assignments and you should be able to find it in whichever version of the book you have.)
- **Take brief notes while reading**. Write your notes on a sheet of paper or on your computer.
- **Answer the questions below**. Write your answers on a sheet of paper or on your computer.
- **Submission:** Take a picture of your notes and answers (or a screenshot, or a pdf of the notes on computer), and upload it to gradescope.
- **Grading:** Grading will be based on effort, not correctness.

2 Questions

Question 1 : 2 points

Provide the notes you took for the readings.

Question 2 : 2 points

For each of the following pairs of functions, determine whether $f(n) \in \mathcal{O}(g(n))$. Give a brief justification for your answer.

(a) $f(n) = 2$ and $g(n) = 5$

(b) $f(n) = 5n$ and $g(n) = 7$

(c) $f(n) = 3n^2 + 7n$ and $g(n) = 2n^2$

Question 3 : 1 point

Prove the following statement using the **inequality definition of** $\mathcal{O}$: $10n^2 = \mathcal{O}(n^3)$